

Assignment 1: Data Pre-processing

Opened: Monday, 3 August 2026, 12:00 AM

Due: Friday, 21 August 2026, 5:30 PM

Assignment Instructions

Extend your Lab 2 notebook into a concise technical report (maximum four pages of written content, excluding the notebook). The report should be written for the Head of Credit Risk at the lending institution, focusing on the quality, reliability, and reproducibility of your machine learning solution.

Your report should include the following sections:

1. Data Quality Audit (5 Marks)

Conduct a comprehensive audit of the dataset and identify the ten columns with the most significant data quality issues (refer to Figure 2.1).

For each column, provide:

- The data quality issue identified.
- The evidence supporting your conclusion (e.g., summary statistics, missing value analysis, duplicate counts, diagnostic plots, or validation checks).
- The missing-data classification where applicable (MCAR, MAR, or MNAR).
- The remediation strategy you selected and why it is appropriate.

Present your findings in a well-organised table.

2. Feature Dictionary (5 Marks)

Create and document at least 15 engineered features drawn from a minimum of four feature families listed in Table 2.2.

For each engineered feature, include:

- Feature name
- Definition (what the feature represents)
- Computation window (the time period over which it is calculated)
- Upstream fields (data lineage) used to construct the feature
- Write time of every upstream field (when each source field becomes available)
- A brief fairness note, where relevant (see Remark 2.3), describing any potential bias or ethical consideration associated with the feature.

3. Leakage Assessment Memo (6 Marks)

Prepare a short technical memo describing the two intentionally planted sources of data leakage.

For each leakage source:

- Identify where it occurs.

- Explain how the leakage happens by referring to the six causes of data leakage discussed in Section 2.7.2.
- Quantify the impact by comparing the model performance before and after the leakage is removed (for example, the change in offline ROC-AUC).
- Recommend a practical process or governance change that would prevent this type of leakage in future model development.

4. Reproducible Machine Learning Pipeline (4 Marks)

Develop a single, end-to-end Scikit-learn **Pipeline** that performs all preprocessing and model training.

Your submission must:

- Perform all preprocessing steps within the pipeline.
- Evaluate the model using group-aware cross-validation.
- Report the final cross-validated ROC-AUC.
- Save the trained pipeline using joblib.
- Ensure the saved pipeline can be reloaded and executed successfully by the marker without modification.

Important Requirements

- Every conclusion or design decision must be supported with evidence, not assumptions.
- Any preprocessing performed outside the cross-validation pipeline will be treated as data leakage and will result in the loss of the pipeline marks, regardless of the model's performance.
- The emphasis of this assignment is on producing a solution that is correct, reproducible, well-documented, and suitable for deployment, rather than simply achieving the highest predictive accuracy.

Source: Strathmore eLearning — Week 2. Data Preparation & Feature Engineering (3–7 Aug)